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Different Boundary Layer Thicknesses

2.Nominal Thickness

3.Displacement Thickness

4.Momentum Thickness

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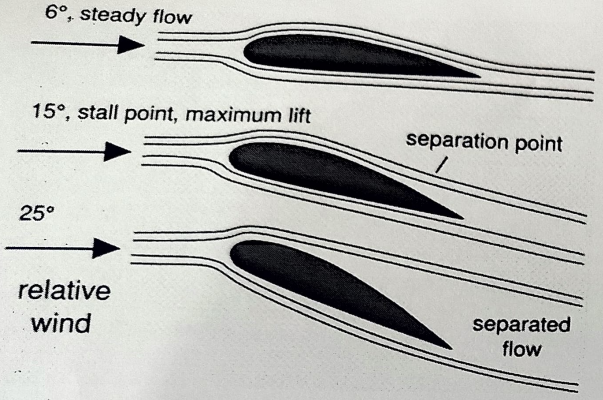
Introduction

- External flows past objects encompass an extremely wide variety of fluid mechanics phenomena. Clearly the character of the flow field is a function of the shape of the body.
- For a given shaped object, the characteristics of the flow depend very strongly on various parameters such as size, orientation, speed, and fluid properties.
- According to dimensional analysis arguments, the character of the flow should depend on the various dimensionless parameters involved.
- For typical external flows the most important of these parameters are the Reynolds number, $Re = UL/v$, where L– is characteristic dimension of the body.

- For many high-Reynolds-number flows the flow field may be divided into two region
  i. A viscous boundary layer adjacent to the surface
  ii. The essentially inviscid flow outside the boundary layer
- W know that fluids adhere the solid walls and they take the solid wall velocity. When the wall does not move also the velocity of fluid on the wall is zero.
- In region near the wall the velocity of fluid particles increases from a value of zero at the wall to the value that corresponds to the external "frictionless" flow outside the boundary layer

Introduction

- The concept of boundary layer was first introduced by a German engineer, Prandtl in 1904.
- According to Prandtl theory, when a real fluid flows past a stationary solid boundary at large values of the Reynolds number, the flow will be divided into two regions.
  i. A thin layer adjoining the solid boundary, called the boundary layer, where the viscous effects and rotation cannot be neglected.
  ii. An outer region away from the surface of the object where the viscous effects are very small and can be neglected. The flow behavior is similar to the upstream flow. In this case a potential flow can be assumed.



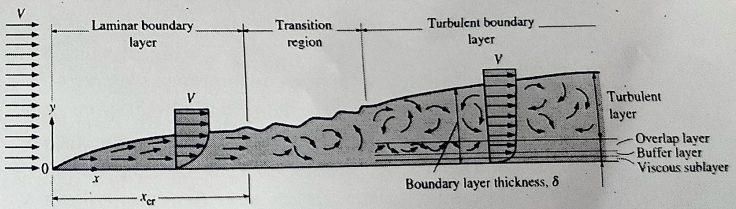
- Since the fluid at the boundaries has zero velocity, there is a steep velocity gradient from the boundary into the flow. This velocity gradient in a real fluid sets up shear forces near the boundary that reduce the flow speed to that of the boundary.
- That fluid layer which has had its velocity affected by the boundary shear is called ***the boundary layer.***
- For smooth upstream boundaries the boundary layer starts out as a ***laminar boundary layer*** in which the fluid particles move in smooth layers.
- As the laminar boundary layer increases in thickness, it becomes unstable and finally transforms into a ***turbulent boundary layer*** in which the fluid particles move in haphazard paths.
- When the boundary layer has become turbulent, there is still a very thon layer next to the boundary layer that has laminar motion. It is called the ***laminar sublayer.***

1.Concept of Boundary Layer:

It is a small region in the immediate vicinity of the boundary surface in which velocity of flowing fluid increases rapidly from zero at the boundary surface to the velocity of main stream of flowing fluid. This layer of flowing fluid is known as BL

i) It is due to the high values of viscosity of flowing fluid in the vicinity of surface.

ii) Due to this effect most of the energy get lost in this zone due to viscous shear effect.



The turbulent bo
f four

- The turbulent boundary layer can be considered to consist of four regions, characterized by the distance from the wall.
- The very thin layer next to the wall where viscous effects are dominant is the **viscous sublayer.** The velocity profile in this layer is very nearly *linear*, and the flow is nearly parallel.
- Next to the viscous sublayer is the **buffer layer,** in which turbulent effects are becoming significant, but the flow is still dominated by viscous effects.
- Above the buffer layer is the **overlap layer,** in which the turbulent effects are much more significant, but still not dominant.
- Above that is the **turbulent** (or **outer**) **layer** in which turbulent effects dominate over viscous effects.

2.Nominal Thickness(δ):

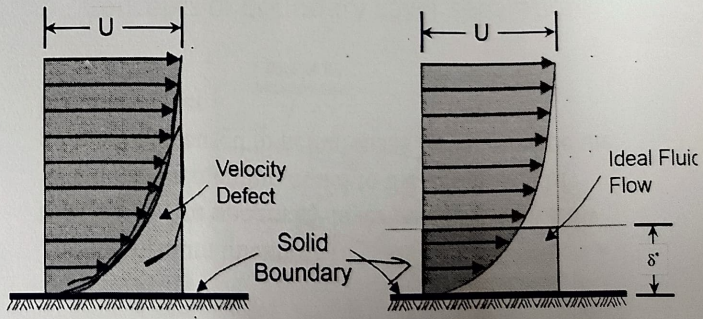
- Nominal thickness of the boundary layer is defined as the thickness of zone extending from solid boundary to a point where velocity reaches 99% of the free stream velocity
- It is based on the fact that beyond this boundary, effect of viscous stresses can be neglected.
- The velocity profile merges smoothly and asymptotically

3.DisplacementThickness(δ*):

✓ Presence of boundary layer introduces a retardation to the free stream velocity in the neighborhood of the boundary.

This causes a decrease in mass flow rate due to presence of boundary layer. A "velocity defect" of (U - u) exists at a certain distance along y axis.

It is Distance (measured perpendicular to the boundary) with which the boundary may be imagined to have been shifted such that the actual flow rate would be the same as that of an ideal fluid (with slip) flowing around the displaced boundary.



Velocity Defect Equivalent Flow Rate

a decrease in mass flow rate due to presence of boundary layer (i.e. due to viscous force) is

$$\int_0^\infty \rho(U-u)b.dy \text{------I}$$

Where,
b is the width of the surface in the direction perpendicular to the flow
ρ is the density of flowing fluid
U is the initial stream velocity of flowing fluid

The mass flow rate deficiency by displacing the solid boundary by δ* will be

$$\rho.U.\delta^*.b \text{------II}$$

Therefore in an incompressible flow equating these two equation will get

$$\rho.U.\delta^*.b = \int_0^\infty \rho(U-u)b.dy$$

$$\therefore \delta^* = \int_0^\infty \left(1-\frac{u}{U}\right).dy$$

Mass flowing per second through the elemental strip is
= ρ x area of strip x velocity

$$= \rho.(b.dy).u$$

Therefore mass momentum of above quantity = mass flowing per second x velocity

$$= \rho.(b.dy).u^2$$

Therefore mass momentum of above quantity in the absence of boundary layer is

$$= (\rho.u.bdy)U$$

Therefore loss of momentum per second is

$$= (\rho.u.bdy)U - \rho.(b.dy).u^2$$
$$= \rho u(U-u)b.dy$$

Therefore total loss of momentum per second is

$$= \int_0^\infty \rho u(U-u)b.dy \text{------I}$$

4.Momentum Thickness(δ^{**}):

Retardation of flow within boundary layer causes a reduction in the momentum flux too.

The momentum thickness (δ**) is defined as the thickness of an imaginary layer in free stream flow which has momentum equal to the deficiency of momentum caused to actual mass flowing inside the boundary layer

But this is also equal to

$$\rho.b.\delta^{**}U^2 \text{-------II}$$

Therefore from equation I and II we have

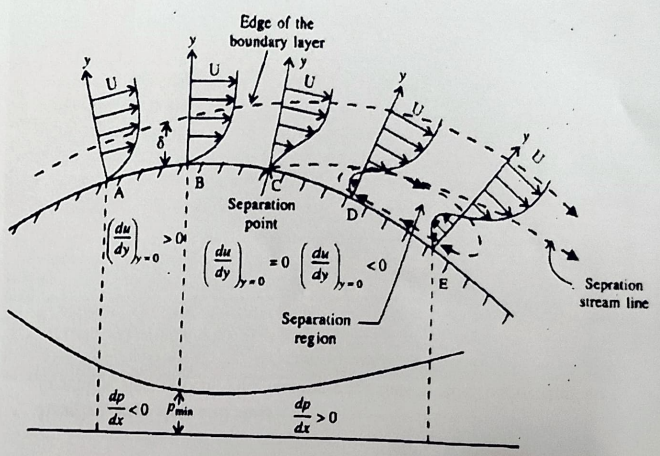
$$\therefore \delta^{**} = \int_0^\infty \frac{u}{U}\left(1-\frac{u}{U}\right).dy$$

5.Energy Thickness(θ):

Energy thickness (θ) is defined as the distance perpendicular to the boundary by which the boundary is to be displaced to compensate for reduction in kinetic energy of fluid caused due to formation of boundary layer. It is given by

$$\therefore \theta = \int_0^\infty \frac{u}{U}\left(1 - \frac{u^2}{U^2}\right).dy$$

Concepts of Boundary Layer separation



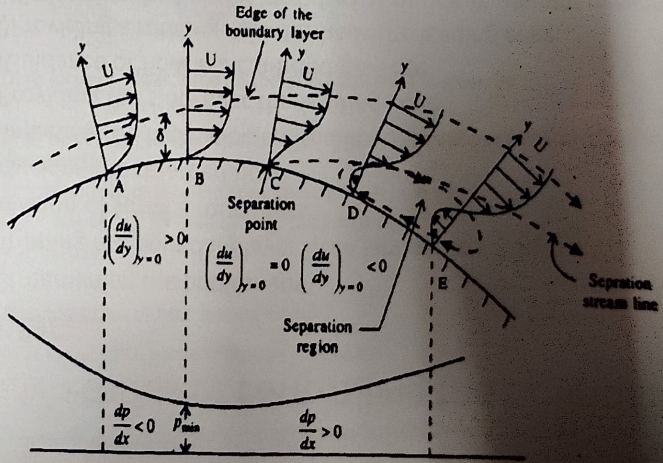
i) When the pressure increases in the direction of flow (dp/dx>0), the pressure forces act opposite to the direction of flow and further increase the retarding effect of the viscous forces.

ii) Subsequently the thickness of the boundary layer increases rapidly in the direction of flow.

iii) If these forces act over a long stretch the boundary layer gets separated from the surface and moves into the main stream.
This phenomenon is called separation.

iv) The point of the body at which the boundary layer is on the verge of separation from the surface is called "point of separation".

Concepts of Boundary Layer separation



v) Figure shows the fluid flows round the surface (the area of flow decreases) it is accelerated over the left hand section until at point B the velocity just outside the boundary is maximum and the pressure is minimum. Thus from A and B the pressure gradient is negative. As long as $dp/dx<0$, the entire boundary layer moves forward.

vi) Beyond B(i.e. along the region BCDE), the area of flow increases and hence velocity of flow decreases; due to decrease of velocity the pressure increases (in the direction of flow) and hence the pressure gradient dp/dx is positive i.e. $dp/dx>0$.

vii) The value of the velocity gradient (du/dy) at the boundary layer is zero at the point C, this point is known as a separation point large turbulent eddies are formed downstream of the point of separation. The disturbed region in which the eddies are formed is called turbulent wake.

7. Methods of controlling the boundary layer:

i)Streamlining the body shape.
ii)Tripping the boundary layer from laminar to turbulent by provision of surface roughness.
iii)Sucking the retarded flow.
iv)Injecting the high velocity fluid in the boundary layer (Slat)
v)Providing the slots near the leading edge.
vi)Guidance of flow in a confined passage.
vii)Providing a rotating cylinder near the leading edge.
viii)Energizing the flow by introducing optimum amount of swirl in the incoming flow

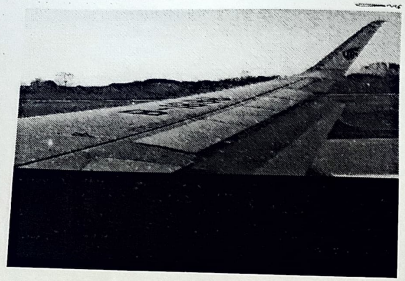
Methods of Boundary-Layer Control in Aerospace Engineering

- **Streamlining of the body/airfoil shape**
  Designing the aircraft body and airfoil with a smooth, streamlined contour to reduce adverse pressure gradients and delay flow separation.
- ***Tripping the boundary layer from laminar to turbulent***
  Introducing controlled surface roughness or a turbulator to promote transition to a turbulent boundary layer, which has greater resistance to separation.
- ***Boundary-layer suction***
  Removing the low-momentum or retarded fluid from the boundary layer through small suction slots or porous surfaces to delay separation.
- ***Boundary-layer blowing / high-velocity air injection***
  Injecting high-momentum air into the boundary layer to increase its energy and prevent or delay flow separation. This principle is used
  blown flaps and circulation-control systems.

- **Leading-edge slots and slats**
  Providing slots or deploying leading-edge slats to allow high-energy air from the lower surface to flow over the upper surface, thereby delaying separation at high angles of attack.
  Guiding the flow through confined passages
  Using suitable aerodynamic surfaces, ducts, or passages to control the direction and distribution of airflow and maintain an attached boundary layer.
  Rotating-cylinder boundary-layer control
  Placing a rotating cylinder near the leading edge to modify the local flow field and increase momentum near the surface, thereby delaying separation.
  Energizing the flow using controlled swirl
  Introducing an optimum amount of swirl into the incoming flow to increase momentum exchange within the boundary layer and improve resistance to flow separation.



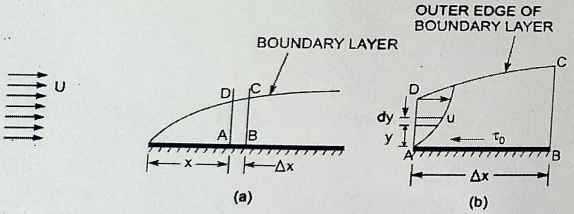






DRAG FORCE ON FLAT PLATE

Due to Boundary Layer



The shear stress is given by

$$\tau = (\mu \frac{\partial u}{\partial y})_{y=0}$$

- Drag force or shear force on the small distance is given by
  $\Delta F_d = \tau \, \Delta x * b$
- Drag force ΔF_d must be equal to the rate of change of momentum over the distance Δx
- The mass rate of flow entering through AD
  $= \int_0^{\delta}$ (ρ× velocity × area of strip thickness dy)
  $= \int_0^{\delta} \rho u b dy$

- The mass rate of flow leaving side BC

mass through AD+$\frac{\partial}{\partial x}$(mass through AD)×$\Delta x$

$$= \int_0^{\delta} \rho u b dy + \frac{\partial}{\partial x}[\int_0^{\delta} \rho u b dy]\Delta x$$

From continuity equation for a steady incompressible flow ,we have

$M_{AD} + M_{DC} = M_{BC}$

$M_{DC} = M_{BC} - M_{AD}$

$= \frac{\partial}{\partial x}\int_0^\delta [\rho u b dy]\Delta x$

momentum flux entering through AD=$\int_0^\delta [\rho u b dy]$x [u]

$= \int_0^\delta \rho u^2 b dy$

Momentum flux entering through side BC = $\int_0^\delta \rho u^2 b dy = \frac{\partial}{\partial x}[\int_0^\delta \rho u^2 b dy]\,\Delta x$

So, momentum flux through DC = $\frac{\partial}{\partial x}[\int_0^\delta \rho u b dy]\,\Delta x$ x U

Rate of change of momentum of control volume = $MF_{BC} - MF_{AD} - MF_{DC}$

$\{\int_0^\delta \rho u^2 b dy + \frac{\partial}{\partial x}[\int_0^\delta \rho u^2 b dy]\,\Delta x\} - \{\int_0^\delta \rho u^2 b dy\} - \{\frac{\partial}{\partial x}[\int_0^\delta \rho u b dy]\,\Delta x$ x u$\}$

$= \rho b \frac{\partial}{\partial x}[\int_0^\delta (u^2 - uU) dy] \times \Delta x$

Total external force is in the direction of rate of change of momentum;

$-\tau \Delta x b = \rho b \frac{\partial}{\partial x}[\int_0^\delta (u^2 - uU) dy] \times \Delta x$

$\frac{\tau}{\rho u^2} = [\frac{\partial}{\partial x}\int_0^\delta \frac{u}{U}(1 - \frac{u}{U})\, dy]$

$$\frac{\tau}{\rho u^2} = \frac{\partial \theta}{\partial x}$$

Where ; θ= Momentum thickness

This is Von karman momentum integral equation

This equation is applicable for all type of boundary layers.

As per momentum principal, total force on the control volume is equal to rate of change of momentum.

IN ABCD, for flat plate P =c so,

$\frac{dP}{dx} = 0$, so there is no external pressure force on control volume ABCD

Only external force is generated due to shear stress, which acts on side AB from B to A

Force is expressed as

$\Delta F_d = \tau\ \Delta x\ b$

In the direction momentum change, force is expressed as,

$\Delta F_d = -\tau\ \Delta x\ b$

Local Co-efficient of Drag [C_D^*]. It is defined as the ratio of the shear stress τ_0 to the quantity $\frac{1}{2}\rho U^2$. It is denoted by C_D^*

Hence $C_D^* = \frac{\tau_0}{\frac{1}{2}\rho U^2}$.

Average Co-efficient of Drag [C_D]. It is defined as the ratio of the total drag force to the quantity $\frac{1}{2}\rho A U^2$. It is also called co-efficient of drag and is denoted by C_D.

Hence $C_D = \frac{F_D}{\frac{1}{2}\rho A U^2}$